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St. Dominic College of Asia

Design and Development of an IoT-Based RFID Attendance, Schedule Validation, and Smart Door Access Control System for Faculty

[an IoT Project]

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Abstract

This study presents the design and development of an IoT-Based RFID Attendance, Schedule Validation, and Smart Door Access Control System for faculty members. The system integrates Radio Frequency Identification (RFID) technology, an ESP32 microcontroller, a real-time clock module, and a web-based management platform to automate attendance tracking, validate teaching schedules, and control classroom access.

The proposed system records faculty attendance through RFID authentication and validates each entry against pre-assigned schedules stored in a centralized database. Upon successful validation, the system activates a relay-controlled solenoid lock to grant access to the classroom. The door remains unlocked during the faculty member's active session and automatically locks upon timeout detection.

The system communicates in real time via Wi-Fi, enabling instant synchronization of attendance logs and access events to a web-based dashboard. It also detects unauthorized access attempts and schedule mismatches, providing alerts for administrative monitoring.

Results from prototype testing demonstrate that the system improves attendance accuracy, enhances classroom security, and reduces manual administrative workload. The study concludes that integrating IoT-based automation with schedule validation significantly contributes to smart campus development and operational efficiency.

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Chapter I

Introduction

The rapid advancement of digital technologies has significantly transformed administrative and security processes in educational institutions. Among these innovations, the Internet of Things (IoT) and Internet of Everything (IoE) enable intelligent automation, real-time monitoring, and seamless integration between hardware and software systems. Educational institutions are increasingly adopting connected technologies to improve operational efficiency, transparency, security, and data accuracy.

This project presents the design and development of an IoT-Based RFID Attendance, Schedule Validation, and Smart Door Access Control System for Faculty. The system automates faculty attendance monitoring, validates assigned teaching schedules, and controls classroom access through RFID authentication integrated with an electronic door locking mechanism.

The proposed system combines RFID technology, ESP32 microcontroller processing, Wi-Fi communication, and a web-based management platform to provide real-time attendance monitoring and smart classroom access control. When a faculty member taps a registered RFID card at the assigned classroom during the correct schedule, the system validates the schedule and automatically unlocks the classroom door using a solenoid lock controlled through a relay module. The door remains unlocked while the faculty member is officially timed-in and automatically locks again once the faculty member times out.

a.1. Overview of IoT/IoE Project

The project is an IoT-enabled attendance and access control solution designed specifically for faculty monitoring and classroom security. The system integrates

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hardware devices, wireless communication, and web technologies to automate attendance logging, schedule validation, and classroom access.

The proposed system:

- Automatically records faculty attendance using RFID technology
- Validates assigned classroom and schedule
- Controls classroom access using a solenoid door lock
- Unlocks the classroom only for authorized faculty members
- Keeps the door unlocked while the faculty member is timed-in
- Automatically locks the classroom after faculty timeout
- Transmits attendance and access logs in real time
- Flags schedule mismatches and unauthorized access attempts

The system follows an IoT architecture where hardware devices communicate with a centralized web-based management platform using wireless networking. The integration of attendance validation and smart locking mechanisms improves both operational efficiency and classroom security.

a.2. Purpose/Objective of the Project

a.2.1. General Objective

To design, develop, and evaluate an IoT-based RFID attendance, schedule validation, and smart door access control system for faculty that enhances accuracy, accountability, security, and administrative efficiency.

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a.2.2. Specific Objectives

1. Implement RFID-based attendance logging for faculty.
2. Develop a schedule validation module to ensure faculty members are present in their assigned rooms.
3. Enable real-time transmission of attendance and access data via Wi-Fi.
4. Integrate IoT-collected data with a web-based management system for real-time updates and reporting.
5. Implement a smart door locking mechanism using a solenoid lock and relay module.
6. Automatically unlock classroom doors only for authorized faculty members assigned to the room and schedule.
7. Automatically relock classroom doors after faculty timeout.
8. Provide alerts or notifications for schedule anomalies or unauthorized access.
9. Evaluate the system's accuracy, reliability, usability, efficiency, and security in a school environment.

a.3. Project Scope and Limitations

a.3.1. Scope

The project includes:

- IoT implementation using **ESP32** microcontroller, **MFRC522** RFID reader, RFID cards, and **DS3231 RTC** module.
- RFID-based **faculty attendance** tracking.
- Real-time **schedule validation**.

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- Smart **classroom access** control using:
    - **Solenoid lock**
    - **5V 1-channel relay module**
    - **12V 5A adapter power supply**
- Automatic **door unlocking** for authorized faculty members.
- Automatic **door locking** after timeout.
- Real-time integration with a web-based **management system**.
- Real-time **alerts or notifications** through the web-based dashboard.

a.3.2. Limitations

- The project is limited to RFID attendance, schedule validation, and door access control.
- The system does not include biometric authentication or facial recognition.
- IoT devices require stable network connectivity for full web integration.
- The solenoid lock mechanism is intended for prototype-level classroom access control.
- Schedule anomaly detection is rule-based and does not yet include predictive AI.
- Only assigned classrooms are monitored by the system.
- During network interruptions, access validation may rely on locally stored configurations.

a.4. Significance and Impact of IoT/IoE Project

For Administration:

- Provides real-time attendance monitoring and classroom access validation.

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- Reduces manual attendance verification and classroom security monitoring.
- Improves transparency and administrative efficiency.
- Enhances classroom access security.

For Faculty:

- Ensures accurate attendance recording.
- Provides automated and secure classroom access.
- Reduces delays caused by manual classroom unlocking procedures.

For the Institution:

- Improves operational efficiency and classroom security.
- Demonstrates institutional innovation through IoT integration.
- Establishes a foundation for future smart campus technologies.
- Supports future AI-based administrative analytics and smart building systems.

a.5. Key Terms and Definitions

Term	Definition
IoT (Internet of Things)	Network of physical devices connected to the internet, collecting and exchanging data.
IoE (Internet of Everything)	Extends IoT to include people, data, processes, and things connected to the network for intelligent interactions

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RFID (Radio Frequency Identification)	Radio Frequency Identification technology used for automatic employee attendance logging.
ESP32	A Wi-Fi-enabled microcontroller used for processing and IoT communication.
Solenoid Lock	Electromagnetic locking mechanism used to automatically lock and unlock classroom doors.
Relay Module	Electrically operated switch used to control the solenoid lock using the ESP32.
Access Control	Restricting or allowing entry to authorized users only.
Schedule Validation	Process of verifying that employees are present in their assigned rooms or shifts at the scheduled time.
Attendance Monitoring	Automatic logging of employee time-in and time-out via IoT devices.
Faculty	Teaching personnel of the school.
Integration	Connection of IoT-collected attendance and validation data with the web-based management system.

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Chapter II

Project Background and Motivation

This chapter discusses the problems in attendance monitoring and classroom access management within educational institutions, reviews existing technologies and their limitations, and explains the motivation behind developing the IoT-Based RFID Attendance, Schedule Validation, and Smart Door Access Control System.

b.1. Problem Statement

Educational institutions commonly experience difficulties in accurately monitoring faculty attendance while also ensuring that classrooms are accessed only by authorized personnel. Traditional attendance systems primarily focus on time logging and often fail to provide real-time classroom access validation.

Existing systems face the following issues:

- Manual attendance processes are prone to errors and delays.
- Most attendance systems do not validate classroom assignments.
- Unauthorized classroom access may occur due to lack of automated security.
- Classroom doors often rely on manual locking and unlocking.
- Attendance systems are commonly isolated from centralized administrative systems.
- Existing systems rarely integrate attendance monitoring with smart door access control.

These challenges highlight the need for a connected system that integrates attendance monitoring, schedule validation, and automated classroom security.

b.2. Existing Solutions and Gaps

b.2.1. RFID-Based Attendance Systems

RFID technology has been widely implemented in attendance systems due to its speed, reliability, and contactless operation. RFID systems reduce manual logging errors and improve automation.

However, many RFID attendance systems only log time-in and time-out records. Most do not validate classroom assignments or control physical classroom access.

b.2.2. Wireless Communication and IoT Integration

IoT-enabled devices such as the ESP32 support wireless communication and real-time monitoring through Wi-Fi connectivity. These technologies allow devices to transmit attendance data to centralized systems instantly.

Despite these advantages, many implementations remain limited to attendance logging and lack integrated classroom security mechanisms.

b.2.3 Smart Access Control Systems

Electronic door locking systems using relay-controlled solenoid locks are commonly used in commercial security systems. These systems provide controlled access through electronic authentication.

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However, most standalone smart lock systems are not integrated with educational attendance systems or schedule validation platforms.

b.2.4. Quality and Evaluation of Software Systems

System quality evaluation is essential for ensuring reliability and usability. The ISO/IEC 25010:2011 standard developed by International Organization for Standardization provides a framework for evaluating software quality characteristics such as functionality, reliability, usability, performance efficiency, and security.

In addition, principles of structured system development discussed by Ruas, Terry and Grosky, William (2020) emphasize the importance of systematic design, integration, testing, and validation to ensure system robustness.

Although standards and engineering principles guide system quality, many institutional attendance systems still operate as isolated hardware tools without deeper validation logic or intelligent monitoring capabilities.

b.2.5. Identified Gaps

The following gaps were identified:

- Existing RFID systems rarely validate room schedules.
- Many attendance systems lack integrated smart door access control.
- Unauthorized access detection is limited.
- Attendance and security systems are usually separated.

- Real-time monitoring and centralized integration are often incomplete.

These gaps justify the development of an integrated IoT-based attendance and smart access control system.

b.3. Project Motivation (Why is your IoE project the right solution?)

The project was developed to provide a unified solution that combines attendance monitoring, schedule validation, and classroom access security.

b.3.1. Complementing the Capstone System

The capstone project focuses on AI-driven administrative management, including schedule optimization, anomaly detection, and clearance compliance. However, AI models require accurate and reliable real-world data.

This IoT project serves as the data acquisition layer of the integrated system architecture:

- The RFID + ESP32 system collects accurate attendance and location data.
- The IoT device transmits real-time records to the web-based platform.
- The web system validates the schedule and flags mismatches.
- The AI module (in the capstone system) analyzes attendance patterns for predictive insights.

Thus, this IoT project provides the physical and communication infrastructure necessary to support intelligent administrative analytics.

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b.3.2. Smart Door Access Workflow

The implemented access control mechanism works as follows:

1. Faculty taps RFID cards.
2. System validates:
    - Faculty identity
    - Assigned classroom
    - Current schedule
3. If validation is successful:
    - Attendance is recorded
    - Relay module is activated
    - Solenoid lock unlocks the classroom door
4. The classroom remains unlocked while the faculty member is officially timed-in.
5. Once the faculty member times out:
    - System deactivates relay module
    - Solenoid lock re-engages
    - Door locks automatically

This workflow improves both convenience and classroom security.

b.3.3. Why IoT-Based RFID is the Right Solution

The system combines:

- RFID-based identification
- Real-time schedule validation

- Automated door access control
- IoT communication
- Centralized web-based monitoring

The addition of a solenoid locking mechanism enhances the system by ensuring that classroom access is granted only to authorized faculty members during their assigned schedules.

b.3.4. Contribution to Institutional Innovation

The project contributes to institutional digital transformation by:

- Enhancing attendance monitoring
- Improving classroom security
- Reducing administrative workload
- Supporting smart campus development
- Establishing scalable IoT infrastructure

Chapter III

System Design and Architecture

This chapter presents the overall system framework, architectural design, components, and communication mechanisms of the proposed IoT-based RFID Attendance and Schedule Validation System.

c.1. System Overview

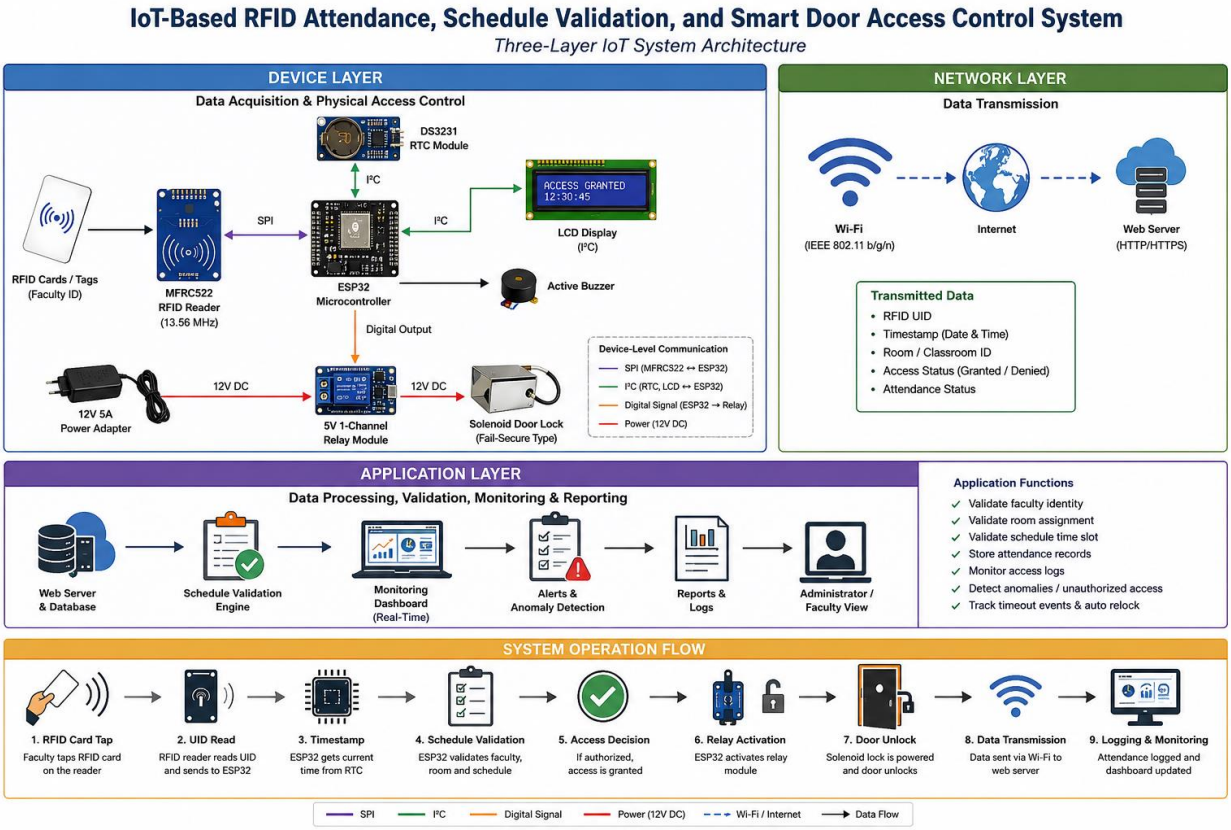


Figure 1: *Three-Layer IoT-Based System Architecture*

The IoT-Based RFID Attendance, Schedule Validation, and Smart Door Access Control System is designed as an integrated hardware and software solution that automates attendance monitoring and classroom access management.

The system follows a three-layer IoT architecture:

c.1.1. Device Layer

The Device Layer includes:

- ESP32 microcontroller
- MFRC522 RFID reader

- RFID cards/tags
- DS3231 RTC module
- 5V 1-channel relay module
- Solenoid door lock
- LCD display
- Active buzzer

When a faculty member taps an RFID card, the RFID reader captures the UID and sends it to the ESP32. The ESP32 retrieves the current time from the RTC module and validates the faculty member's assigned room and schedule.

If validation succeeds, the ESP32 activates the relay module which powers the solenoid lock and unlocks the classroom door.

c.1.2. Network Layer

The Network Layer uses Wi-Fi communication through the ESP32 to transmit attendance and access control data to the web server using HTTP/HTTPS protocols.

c.1.3. Application Layer

The Application Layer consists of the web-based management system and database. The system:

- Stores attendance records
- Validates schedules
- Monitors access logs

- Displays real-time monitoring data
- Flags anomalies and unauthorized access attempts

The system also tracks faculty timeout events to automatically relock classroom doors.

c.2. Components Used (Devices, Sensors, etc.)

c.2.1. Processing and Communication

c.2.1.1. ESP32 Development Board

- Main microcontroller of the system
- Processes RFID and schedule validation logic
- Controls relay module and solenoid lock
- Provides Wi-Fi connectivity

c.2.2. Identification

c.2.2.1. MFRC522 RFID Reader Module

- Operates at 13.56 MHz
- Reads RFID cards contactlessly
- Communicates with ESP32 using SPI protocol

c.2.2.2. RFID Cards/Tags

- Issued to faculty members
- Contain unique identification numbers

c.2.3. Monitoring and Display

c.2.3.1. LCD Display

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- Displays attendance confirmation
- Displays access granted or denied messages
- Displays system status information

c.2.3.2 Computer Workstation

- Hosts web-based monitoring dashboard
- Displays attendance records and access logs

c.2.4. Time Management

c.2.4.1. DS3231 RTC Module

- Maintains accurate date and time
- Provides reliable timestamping

c.2.5. Smart Access Control Components

c.2.5.1. Solenoid Door Lock

- Controls physical classroom door locking mechanism
- Unlocks when authorized faculty access is granted
- Locks automatically after timeout

c.2.5.2. 5V 1-Channel Relay Module

- Acts as an electronic switch controlled by the ESP32
- Controls power delivery to the solenoid lock
- Isolates high-voltage load from the microcontroller

c.2.5.3. 12V 5A Adapter Power Supply

- Provides stable external power for the solenoid lock

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- Ensures reliable operation of relay-controlled locking mechanism

c.3. Communication and Networking

c.3.1. Device-Level Communication

The MFRC522 RFID reader communicates with the ESP32 using the SPI protocol. The DS3231 RTC module and LCD display communicate through the I2C protocol.

The relay module is connected to a digital output pin of the ESP32. When schedule validation succeeds, the ESP32 sends a HIGH or LOW signal to activate the relay and control the solenoid lock.

c.3.2. Network-Level Communication

The ESP32 transmits attendance and access control data to the web server using Wi-Fi and HTTP/HTTPS requests.

The transmitted data includes:

- RFID UID
- Timestamp
- Room identifier
- Access status
- Attendance status

c.3.3. Application-Level Communication

c.3.3.1. The web server validates:

- Faculty identity

- Classroom assignment
- Schedule time slot

c.3.3.2. If authorized:

- Attendance is logged
- Access is granted
- Dashboard is updated

c.3.3.3. If unauthorized:

- Access is denied
- Event is logged as an anomaly
- Door remains locked

c.3.4. Overall Communication Flow

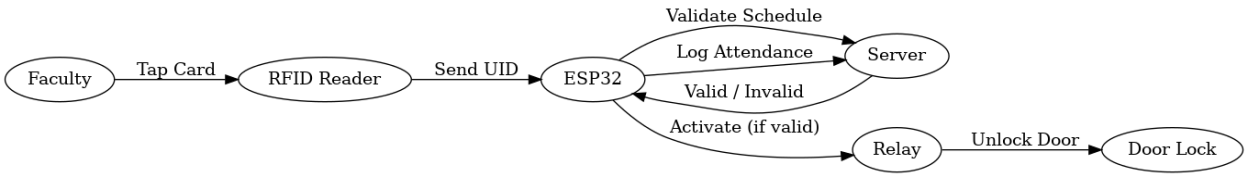


Figure 2: *Sequence Diagram of IoT*

RFID Card → RFID Reader → ESP32 → RTC Timestamp → Schedule Validation → Relay Activation → Solenoid Lock Control → Wi-Fi Data Transmission → Web Server → Dashboard Monitoring

c.4. Entity Relationship Diagram

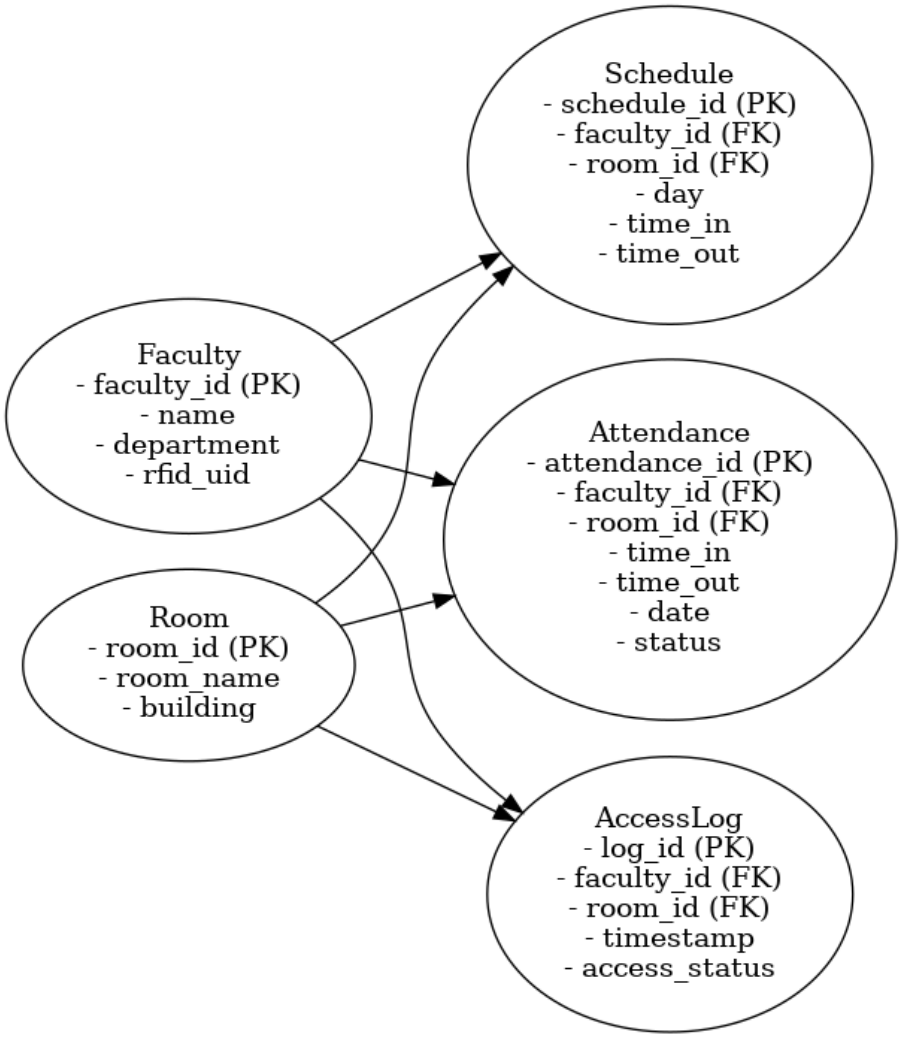


Figure 3: *Entity Relationship Diagram of RFID device*

The Entity-Relationship Diagram (ERD) illustrates the logical structure of the database used in the IoT-Based RFID Attendance, Schedule Validation, and Smart Door Access Control System. It defines the entities involved in the system, their attributes, and the relationships between them.

The system is composed of five primary entities: **Faculty, Schedule, Room, Attendance, and AccessLog**. The **Faculty** entity stores the personal and identification

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details of each faculty member, including their RFID UID, which serves as the primary identifier for system authentication.

The **Schedule** entity contains the assigned teaching schedules of faculty members, including the room allocation, day, and time-in/time-out information. This entity is directly linked to both the Faculty and Room entities, ensuring that each faculty member is validated based on their assigned classroom and schedule.

The **Room** entity represents the physical classrooms being monitored and controlled by the system. It serves as a reference point for both scheduling and access validation.

The **Attendance** entity records the time-in and time-out logs of faculty members after successful RFID authentication and schedule validation. It provides a historical record of attendance activity per faculty and room.

The **AccessLog** entity captures all RFID access attempts, including both successful and unsuccessful entries. It stores timestamped records of access status, enabling the system to detect unauthorized attempts and anomalies.

Overall, the relationships between entities ensure data consistency and enable real-time validation of attendance, scheduling, and access control operations within the IoT system.

c.5. Use Case Diagram

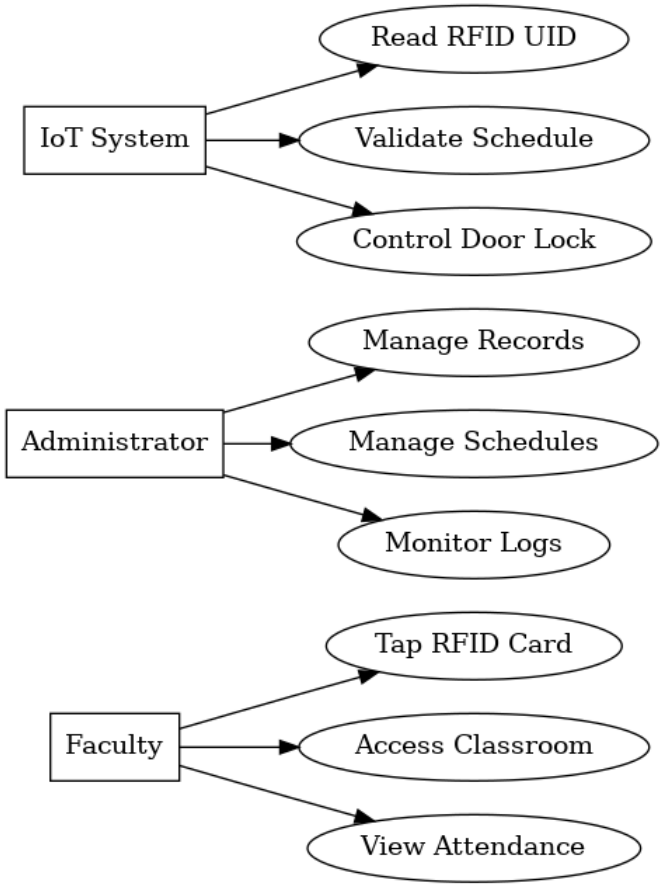


Figure 4: *Use Case Diagram of RFID device*

The Use Case Diagram represents the functional interactions between the system actors and the IoT-Based RFID Attendance and Smart Door Access Control System. It identifies the roles of each user and how they interact with the system's functionalities.

There are three primary actors in the system: **Faculty, Administrator, and IoT System (ESP32-based device)**.

The **Faculty** actor interacts directly with the system by tapping their RFID card to access classrooms, triggering the authentication process. Faculty members can also view their attendance records through the web-based dashboard.

The **Administrator** is responsible for managing the system's administrative functions. This includes managing faculty profiles, configuring and updating schedules, and monitoring attendance and access logs. The administrator ensures that all data in the system remains accurate and properly maintained.

The **IoT System** acts as the core processing unit that bridges hardware and software components. It is responsible for reading RFID data, validating schedules through server communication, recording attendance, and controlling the solenoid door lock via the relay module.

The use cases are interconnected in a workflow where RFID scanning initiates the process, followed by schedule validation and access decision-making. If validation is successful, the system grants access and records attendance; otherwise, it denies entry and logs the attempt as an anomaly.

Overall, the Use Case Diagram demonstrates how the system automates attendance monitoring and access control while maintaining security, accuracy, and administrative efficiency.

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Chapter IV

Hardware and Software Implementation

d.1. Hardware Components (e.g., sensors, microcontrollers)

The system is composed of several hardware components that work together to perform automated identification and display functions. The main components used in the project are the following:

d.1.1. ESP32 microcontroller

Serves as the main processing unit of the system. It is responsible for controlling all connected devices, processing data from the RFID module, and sending output to the display. The ESP32 is chosen because of its fast processing speed, built-in Wi-Fi capability, and multiple GPIO pins.

d.1.2. MFRC522 RFID module

Used to read RFID cards or tags. It communicates with the ESP32 through the SPI (Serial Peripheral Interface) protocol. When a card is placed near the module, it reads the unique identifier (UID) which is then processed by the system.

d.1.3. RFID cards/tags

Are used as input devices. Each card contains a unique UID that serves as the identity of the user in the system.

d.1.4. I2C LCD display (16x2)

Used to display information such as the scanned card UID or system messages. It uses the I2C communication protocol, which reduces the number of wires needed for connection.

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d.1.5. DS3231 Real-Time Clock (RTC) module

Used to provide accurate date and time information. It communicates with the ESP32 using the I2C protocol. This component ensures that every RFID scan is recorded with a precise timestamp, even if the system is restarted.

d.1.6. Active buzzer

Used as an output device to provide audio feedback. It produces a sound when a card is successfully scanned or when an error occurs, enhancing user interaction with the system.

d.1.7. Breadboard and Jumper Wires

Are used to connect all the components without soldering, allowing easy modification and testing of the circuit.

d.1.8. Solenoid Lock

Functions as the electronic locking mechanism for classroom access control.

d.1.9. 5V 1-Channel Relay Module

Controls the switching of power to the solenoid lock based on commands from the ESP32.

d.1.10. 12V 5A Adapter Power Supply

Provides stable and sufficient power for the relay module and solenoid lock.

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d.2. Software/Platform Used

The system is developed using software tools and libraries that enable communication and control of hardware components.

d.2.1. Arduino IDE (Integrated Development Environment)

Used to write, compile, and upload the program to the ESP32 microcontroller. It provides a simple and efficient platform for embedded system development.

The system is programmed using the **C/C++ programming language**, which allows precise control of hardware and system logic.

The following libraries are utilized:

d.2.1.1. MFRC522 Library

Enables communication between the ESP32 and RFID module.

d.2.1.2. SPI Library

Handles SPI communication for RFID data transfer.

d.2.1.3. Wire Library

Supports I2C communication for both LCD and RTC modules.

d.2.1.4. LiquidCrystal_I2C Library

Used to control and display text on the LCD.

d.2.1.5. RTClib (DS3231 Library)

Used to interface with the DS3231 RTC module and retrieve accurate date and time.

These libraries simplify coding and ensure reliable interaction between hardware components.

d.3. System Integration

d.3.1. When the system starts:

1. ESP32 initializes all modules.
2. LCD displays readiness status.
3. ESP32 connects to Wi-Fi.
4. The system waits for an RFID scan.

d.3.2. When a faculty member taps an RFID card:

1. RFID UID is read.
2. Current timestamp is retrieved.
3. Data is validated against assigned room and schedule.

d.3.3. If validation succeeds:

- Attendance is recorded
- Relay module activates
- Solenoid lock unlocks the door
- LCD displays “Access Granted”
- Buzzer provides confirmation sound

d.3.4. If validation fails:

- Door remains locked
- LCD displays “Access Denied”

- Error event is logged
- Buzzer alerts user

d.3.5. Once the faculty member times out:

- Attendance timeout is recorded
- Relay deactivates
- Solenoid lock locks the door automatically

This integration creates a secure and automated attendance and access control environment.

Chapter V

Project Development and Implementation Process

This chapter presents the systematic process followed in the design, development, testing, and deployment of the IoT-Based RFID Attendance and Schedule Validation System. It describes how the project was planned, implemented, and validated to ensure functionality, reliability, and integration with the web-based management system.

e.1. Project Planning and Design

The system was designed to combine attendance monitoring and smart access control into a single IoT-based solution.

e.1.1. The planning phase included:

- Requirement analysis
- Hardware selection
- IoT architecture design

- Database planning
- API design
- Access control workflow design

e.1.2. The implemented workflow is:

1. Faculty taps RFID card
2. UID is scanned
3. Timestamp is generated
4. Server validates schedule
5. Relay activates if authorized
6. Solenoid lock unlocks classroom door
7. Door relocks after timeout

e.2. Project Development Phases

e.2.1. Hardware Development Phase

Included assembly and testing of:

- ESP32
- RFID reader
- RTC module
- Relay module
- Solenoid lock
- LCD display
- Buzzer

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e.2.2. Firmware Development Phase

Included programming:

- RFID reading
- Wi-Fi communication
- Schedule validation logic
- Relay control
- Door locking logic

e.2.3. Web Integration Phase

Included API communication and dashboard integration.

e.2.4. Testing and Debugging Phase

Each module was individually tested before full integration.

e.2.5. Deployment Phase

Prototype-level deployment was conducted in a controlled environment.

e.3. Hardware Setup and Installation

e.3.1. Component Connections

e.3.1.1. RFID Module (MFRC522) via SPI

- SDA → D5
- SCK → D18
- MOSI → D23
- MISO → D19
- RST → D4

c.3.1.2. RTC Module (DS3231) via I2C

- SDA → D21
- SCL → D22

e.3.1.3. LCD Display (I2C)

- SDA → D21
- SCL → D22

e.3.1.4. Relay Module

- IN → D26
- VCC → 5V
- GND → GND

e.3.1.5. Solenoid Lock

- Connected to relay module
- Powered by 12V 5A adapter

e.3.2. Physical Installation

- Components enclosed in protective casing
- RFID reader mounted near classroom entrance
- Solenoid lock attached to classroom door
- Power supply securely installed

e.4. Software Development and Configuration

e.4.1. Embedded System Development

Firmware functionalities include:

- RFID card scanning
- Schedule validation
- Door unlock and lock control
- Wi-Fi communication
- Attendance logging
- Error handling

e.4.2. Data Transmission Format

The ESP32 transmits data using JSON format containing:

- RFID UID
- Timestamp
- Room ID
- Access status
- Attendance status

e.4.3. Web System Configuration

The server performs:

- Faculty verification
- Schedule validation
- Attendance logging
- Access control logging

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- Real-time monitoring

e.4.4. User Interface

The web dashboard displays:

- Attendance logs
- Access control records
- Unauthorized access attempts
- Real-time room monitoring

e.5. System Testing and Validation

e.5.1. Functional Testing

Tested:

- RFID scanning
- RTC accuracy
- Relay switching
- Solenoid lock operation
- Wi-Fi communication
- API connectivity

e.5.2. Integration Testing

Verified:

- End-to-end data flow
- Door unlocking functionality
- Automatic relocking
- Dashboard updates

e.5.3. Performance Testing

Evaluated:

- Scan response time
- Lock activation speed
- Network reliability
- Attendance validation accuracy

e.5.4. Validation Criteria

System evaluated based on:

- Accuracy
- Reliability
- Security
- Efficiency
- Usability

e.6. Deployment and Integration

e.6.1. System Deployment

RFID stations with smart locking mechanisms are intended to be installed at classroom entrances.

Each station includes:

- ESP32
- RFID reader
- RTC module
- Relay module

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- Solenoid lock
- Power supply

e.6.2. System Integration

The system integrates:

- Attendance monitoring
- Schedule validation
- Smart access control
- Web-based monitoring

e.6.3. Operational Workflow After Deployment

1. Faculty taps RFID cards.
2. RFID reader captures UID.
3. ESP32 retrieves timestamps.
4. Server validates room and schedule.
5. If authorized:
    - Attendance recorded
    - Relay activated
    - Door unlocked
6. The faculty conducts class.
7. Upon timeout:
    - Attendance timeout recorded
    - Relay deactivated
    - Door locked automatically

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e.6.4 Current Deployment Status

Prototype-level integration testing confirms:

- RFID scanning functionality
- Relay activation functionality
- Solenoid lock operation
- Wi-Fi communication
- Basic API communication
- Automated door locking and unlocking

Full-scale deployment remains for future implementation.

Chapter VI

Results and Findings

f.1. System Performance

Initial testing indicates that the system successfully performs:

- RFID card detection
- Accurate timestamp generation
- Real-time Wi-Fi communication
- Attendance recording
- Schedule validation
- Automated door unlocking
- Automatic door relocking after timeout

The **relay module** successfully controls the solenoid lock based on access validation results.

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The **12V 5A power supply** provides stable operation for the locking mechanism during continuous testing.

Preliminary testing also confirms that **unauthorized RFID cards** do not unlock the classroom door.

f.2. Challenges Faced and How They Were Solved

f.2.1. RFID Recognition Issues

Some RFID cards were initially not recognized.

Solution: UID registration and database mapping were standardized.

f.2.2. Wi-Fi Connectivity Instability

ESP32 occasionally disconnects from Wi-Fi.

Solution: Automatic reconnection logic was implemented.

f.2.3. Power Supply Fluctuations

The solenoid lock required higher power than the ESP32 could directly provide.

Solution: A separate 12V 5A adapter power supply and relay module were implemented.

f.2.4. Relay Switching Issues

Incorrect relay triggering initially caused inconsistent lock behavior.

Solution: Relay control logic and wiring were corrected and tested.

f.2.5. Solenoid Lock Heating

Extended continuous activation caused slight heating.

Solution: Proper activation timing and stable power regulation were implemented.

f.2.6. Integration Complexity

Synchronizing attendance validation with access control required careful system coordination.

Solution: Structured API communication and modular programming were implemented.

Despite these challenges, the system achieved successful prototype-level operation.

Chapter VII

Conclusion

g.1. Summary of the Project

The IoT-Based RFID Attendance and Schedule Validation System for Faculty aims to address the limitations of traditional attendance systems by introducing automation, real-time monitoring, and schedule validation through IoT technology.

The system integrates RFID-based identification, ESP32 microcontroller processing, real-time clock timestamping, and web-based validation to create a unified attendance monitoring solution. It is designed to ensure that faculty attendance is not only recorded but also verified against assigned schedules and classroom locations.

Although the system is still under development, initial testing has demonstrated the feasibility of integrating hardware and software components into a functional IoT-based

solution. The project establishes a strong foundation for improving accuracy, transparency, and administrative efficiency in educational institutions.

g.2. Future Work and Enhancements (Recommendations)

To further improve the system, the following enhancements are recommended:

- **Full System Deployment and Testing**
  Conduct comprehensive real-world testing across multiple classrooms to evaluate performance under actual conditions.
- **AI-Based Anomaly Detection**
  Integrate machine learning algorithms to analyze attendance patterns and detect unusual behaviors automatically.
- **Mobile Notification System**
  Implement SMS or mobile app notifications for real-time alerts on schedule mismatches.
- **Offline Data Handling**
  Add local data storage to allow the system to function during network interruptions and sync data once connectivity is restored.
- **User Interface Enhancement**
  Improve the web dashboard for better visualization, reporting, and usability.
- **Multi-Room Scalability**
  Expand the system to support multiple RFID stations across different buildings.

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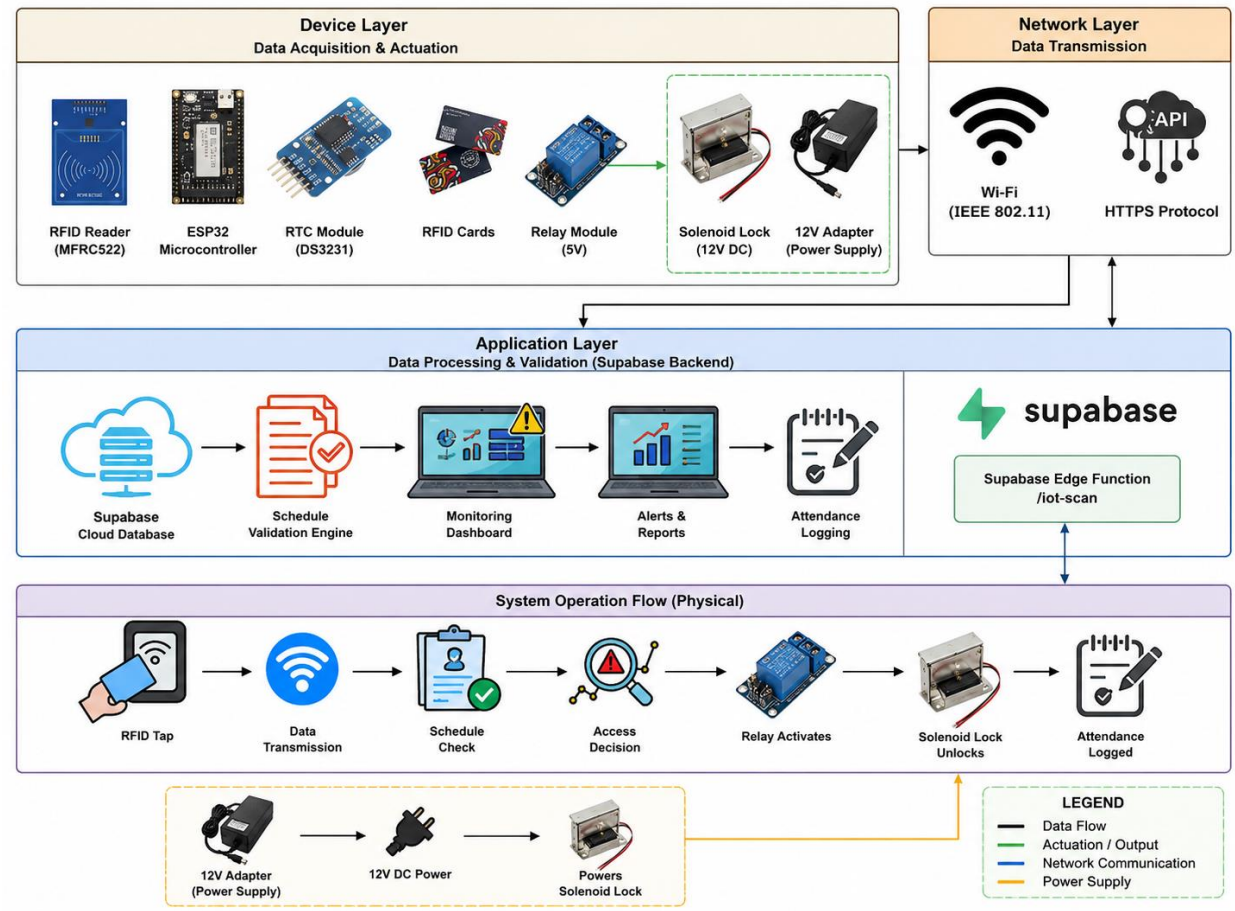
https://www.researchgate.net/publication/365946272_Software_Engineering_A_Practitioner's_Approach_9_th_Edition

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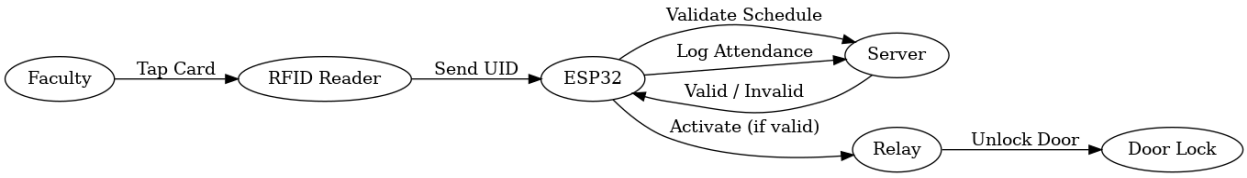
Appendices

Appendix A - Diagrams, Schematics, or Screenshots

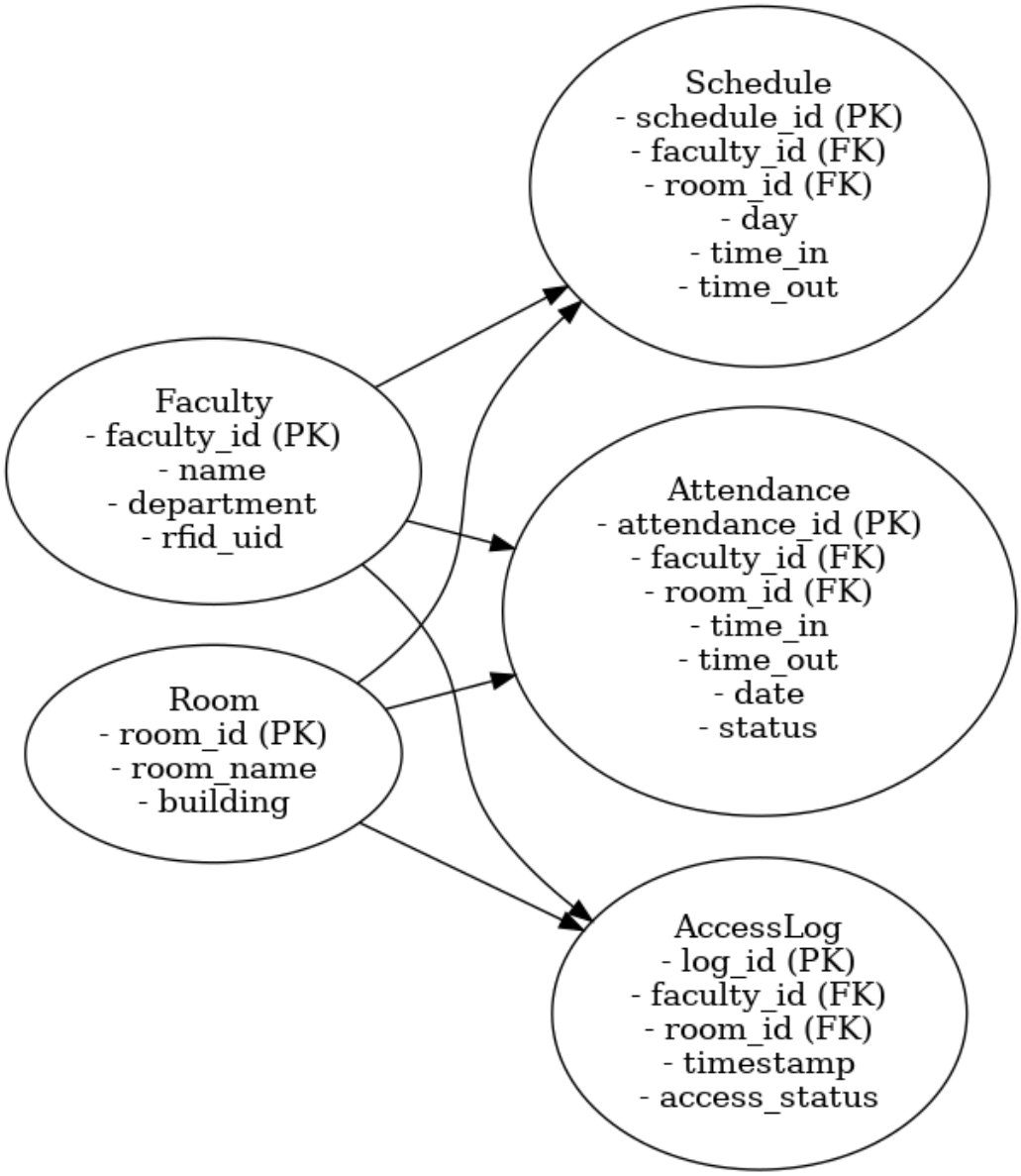
A.1. System Architecture



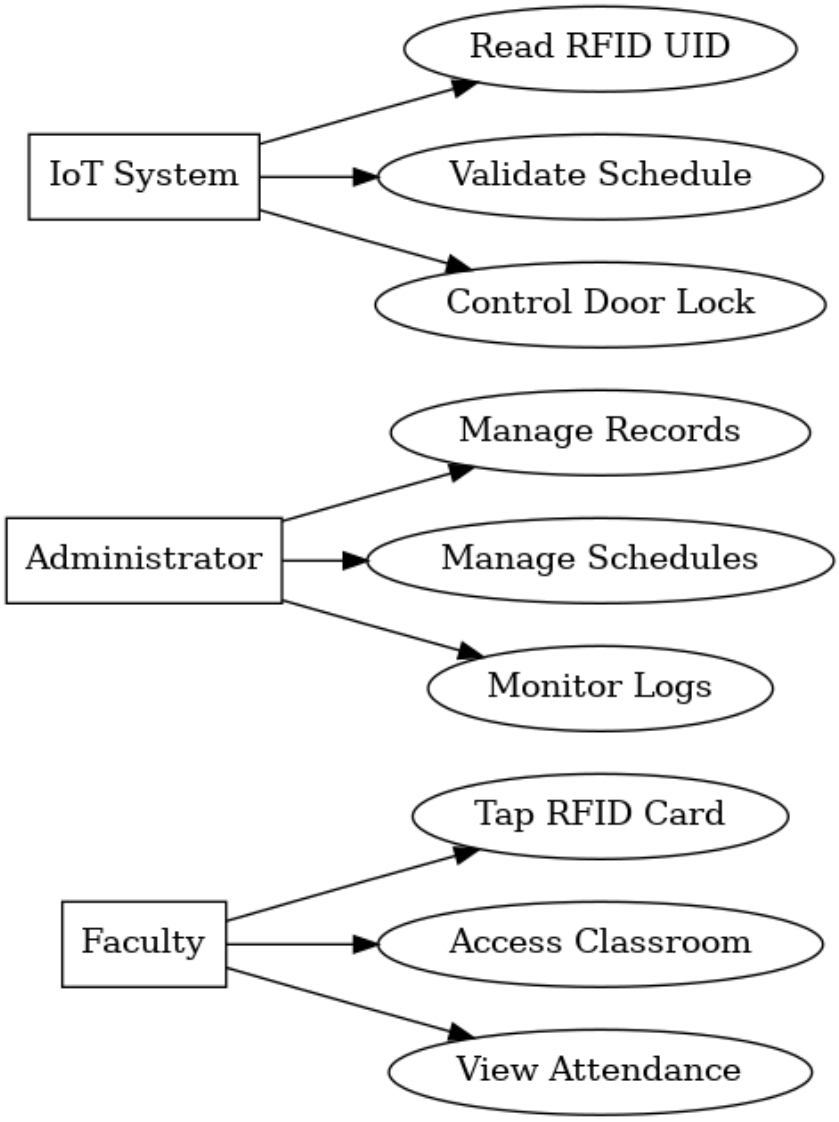
A.2. Sequence Diagram



A.3. Entity Relationship Diagram



A.4. Use Case Diagram



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Appendix B - Codes or Scripts Used

B.1. Computer Laboratory 1 Code

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include "RTClib.h"
#include <LiquidCrystal_I2C.h>
#include <ArduinoJson.h>
#include <time.h>

// ================ WIFI ================
const char* ssids[] = {
  "BOOM",
  "GlobeAtHome_5DD78_2.4"
};

const char* passwords[] = {
  "@M4xRT&VDSL",
  "025C300D"
};

const int wifiCount = 2;

int currentWiFi = 0;

unsigned long lastWiFiRetry = 0;
const unsigned long wifiRetryInterval = 10000;
```

```
// ================ SERVER ================
const char* SERVER_URL =
"https://ferocious-penholder-blaming.ngrok-free.dev/api/iot/scans";

// ================ ROOM ================
const char* DEVICE_ID = "Computer-Laboratory-1";
const char* ROOM_NAME = "Computer Lab 1";

// ================ RELAY STATE ================
#define LOCK_STATE LOW
#define UNLOCK_STATE HIGH

// ================ HARDWARE ================
#define SS_PIN 5
#define RST_PIN 4

MFRC522 rfid(SS_PIN, RST_PIN);

RTC_DS3231 rtc;
LiquidCrystal_I2C lcd(0x27, 16, 2);

#define BUZZER 25
#define RELAY_PIN 26

unsigned long lastUpdate = 0;
```

```
// ================ BUZZER SOUNDS ================
void beepSuccess() {

  tone(BUZZER, 1200);
  delay(120);
  noTone(BUZZER);

  delay(80);

  tone(BUZZER, 1800);
  delay(120);
  noTone(BUZZER);
}

void beepDenied() {

  for (int i = 0; i < 3; i++) {

    tone(BUZZER, 300);
    delay(120);
    noTone(BUZZER);

    delay(80);
  }
}
```

```
void beepError() {

  for (int i = 0; i < 2; i++) {

    tone(BUZZER, 150);
    delay(400);
    noTone(BUZZER);

    delay(150);
  }
}
```

```
// ================ HOME SCREEN ================
void showHomeScreen() {

  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print(ROOM_NAME);

  lastUpdate = 0;
}

// ================ RTC ================
void setupRTC() {

  if (!rtc.begin()) {

    lcd.clear();
    lcd.print("RTC Failed");

    while (1);
  }
```

```
  // AUTO SET RTC IF BATTERY LOST
  if (rtc.lostPower()) {

    rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));
  }
}
```

```
// ================ CLOCK ================
void updateClock() {

  if (millis() - lastUpdate >= 1000) {

    lastUpdate = millis();

    DateTime now = rtc.now();

    char line[17];

    sprintf(line,
            "%02d:%02d:%02d",
            now.hour(),
            now.minute(),
            now.second());

    lcd.setCursor(0, 1);
    lcd.print("                ");

    lcd.setCursor(0, 1);
    lcd.print(line);
  }
}
```

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```
// ================ DOOR ================
void openDoor() {

  digitalWrite(RELAY_PIN, UNLOCK_STATE);

  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print("Door Open");

  beepSuccess();

  delay(5000);

  digitalWrite(RELAY_PIN, LOCK_STATE);

  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print("Door Locked");

  delay(1000);

  showHomeScreen();
}
```

```
// ================ WIFI ================
void connectWiFi() {

  if (WiFi.status() == WL_CONNECTED)
    return;

  // PREVENT RECONNECT SPAM
  if (millis() - lastWiFiRetry < wifiRetryInterval)
    return;

  lastWiFiRetry = millis();

  lcd.clear();
  lcd.print("Connecting WiFi");

  Serial.println("\nConnecting WiFi...");

  WiFi.mode(WIFI_STA);

  // IMPORTANT
  WiFi.setSleep(false);

  // REMOVE OLD CONNECTION
  WiFi.disconnect();

  delay(500);
```

```
  WiFi.begin(
    ssids[currentWiFi],
    passwords[currentWiFi]
  );

  int attempts = 0;

  while (WiFi.status() != WL_CONNECTED &&
         attempts < 20) {

    delay(500);

    Serial.print(".");

    attempts++;
  }

  if (WiFi.status() == WL_CONNECTED) {

    Serial.println("\nWiFi Connected!");

    lcd.clear();
    lcd.print("WiFi Connected");

    beepSuccess();
```

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```
    delay(1000);

    showHomeScreen();

    return;
  }

  Serial.println("\nWiFi Failed");

  lcd.clear();
  lcd.print("WiFi Failed");

  beepError();

  // SWITCH TO NEXT WIFI
  currentWiFi++;

  if (currentWiFi >= wifiCount)
    currentWiFi = 0;
}
```

```
// ================ RFID PROCESS ================
void sendRFIDScan(String uid) {

  beepSuccess();

  DateTime now = rtc.now();

  char timeStr[9];

  sprintf(timeStr,
          "%02d:%02d:%02d",
          now.hour(),
          now.minute(),
          now.second());

  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print("UID:");
  lcd.print(uid.substring(0, 8));

  lcd.setCursor(0, 1);
  lcd.print(timeStr);

  delay(1500);
```

```
  // CHECK WIFI
  if (WiFi.status() != WL_CONNECTED) {

    lcd.clear();
    lcd.print("WiFi Offline");

    beepError();

    delay(1500);

    showHomeScreen();

    return;
  }

  HTTPClient http;

  String payload = "{";
  payload += "\"deviceId\":\"" + String(DEVICE_ID) + "\",";
  payload += "\"uid\":\"" + uid + "\",";
  payload += "\"status\":\"success\"";
  payload += "}";

  Serial.println("\nSending Request...");
  Serial.println(payload);
```

```
299   http.begin(SERVER_URL);
300
301   http.addHeader("Content-Type", "application/json");
302
303   int code = http.POST(payload);
304
305   Serial.print("HTTP CODE: ");
306   Serial.println(code);
307
308   if (code <= 0) {
309
310     lcd.clear();
311     lcd.print("Server Error");
312
313     beepError();
314
315     delay(1500);
316
317     showHomeScreen();
318
319     http.end();
320
321     return;
322   }
```

```
324   String response = http.getString();
325
326   Serial.println(response);
327
328   http.end();
329
330   StaticJsonDocument<2048> doc;
331
332   DeserializationError error =
333     deserializeJson(doc, response);
334
335   if (error) {
336
337     lcd.clear();
338     lcd.print("Parse Error");
339
340     beepError();
341
342     delay(1500);
343
344     showHomeScreen();
345
346     return;
347   }
```

```
349   bool ok = doc["success"] | false;
350
351   String scanStatus =
352     doc["scan"]["status"] | "";
353
354   String analysis =
355     doc["analysis"]["status"] | "";
356
357   String action =
358     doc["attendance"]["action"] | "";
359
360   String name =
361     doc["user"]["full_name"] | "Unknown";
362
363   lcd.clear();
```

```
  // ACCESS GRANTED
  if (ok &&
      scanStatus == "success" &&
      (analysis == "on_time" ||
       analysis == "late")) {

    lcd.setCursor(0, 0);

    if (action == "TIME-IN")
      lcd.print("TIME IN");

    else if (action == "TIME-OUT")
      lcd.print("TIME OUT");

    else
      lcd.print("ACCESS OK");

    lcd.setCursor(0, 1);
    lcd.print(name.substring(0, 16));

    delay(1500);

    openDoor();

    return;
  }
```

```
  // ACCESS DENIED STATES
  else if (analysis == "wrong_room") {

    lcd.print("Wrong Room");
  }

  else if (analysis == "outside_schedule") {

    lcd.print("Wrong Schedule");
  }

  else if (analysis == "no_schedule") {

    lcd.print("No Schedule");
  }

  else if (scanStatus == "not_registered") {

    lcd.print("Card Unknown");
  }

  else {

    lcd.print("Access Denied");
  }
```

```
  beepDenied();

  delay(1500);

  showHomeScreen();
}
```

```
// ================ SETUP ================
void setup() {

  Serial.begin(115200);

  SPI.begin();

  rfid.PCD_Init();

  Wire.begin(21, 22);

  lcd.init();
  lcd.backlight();

  pinMode(BUZZER, OUTPUT);

  pinMode(RELAY_PIN, OUTPUT);

  digitalWrite(RELAY_PIN, LOCK_STATE);

  setupRTC();

  beepSuccess();

  connectWiFi();

  showHomeScreen();
}
```

```
// ================ LOOP ================
void loop() {

  // AUTO WIFI RECONNECT
  if (WiFi.status() != WL_CONNECTED) {

    connectWiFi();
  }

  // RFID SCAN
  if (rfid.PICC_IsNewCardPresent() &&
      rfid.PICC_ReadCardSerial()) {

    String uid = "";

    for (byte i = 0; i < rfid.uid.size; i++) {

      if (rfid.uid.uidByte[i] < 0x10)
        uid += "0";

      uid += String(
               rfid.uid.uidByte[i],
               HEX
             );
    }
```

```
    uid.toUpperCase();

    sendRFIDScan(uid);

    rfid.PICC_HaltA();
    rfid.PCD_StopCrypto1();
  }

  updateClock();

  delay(100);
}
```

B.2. Room 312 Code

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include "RTClib.h"
#include <LiquidCrystal_I2C.h>
#include <ArduinoJson.h>
#include <time.h>

// ================ WIFI ================
const char* ssids[] = {
  "BOOM",
  "GlobeAtHome_5DD78_2.4"
};

const char* passwords[] = {
  "@M4xRT&VDSL",
  "025C300D"
};

const int wifiCount = 2;

int currentWiFi = 0;

unsigned long lastWiFiRetry = 0;
const unsigned long wifiRetryInterval = 10000;
```

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```
// ================ SERVER ================
const char* SERVER_URL =
"https://ferocious-penholder-blaming.ngrok-free.dev/api/iot/scans";

// ================ ROOM ================
const char* DEVICE_ID = "Room-312";
const char* ROOM_NAME = "Room 312";

// ================ RELAY STATE ================
#define LOCK_STATE LOW
#define UNLOCK_STATE HIGH

// ================ HARDWARE ================
#define SS_PIN 5
#define RST_PIN 4

MFRC522 rfid(SS_PIN, RST_PIN);

RTC_DS3231 rtc;
LiquidCrystal_I2C lcd(0x27, 16, 2);

#define BUZZER 25
#define RELAY_PIN 26

unsigned long lastUpdate = 0;
```

```
// ================ BUZZER SOUNDS ================
void beepSuccess() {

  tone(BUZZER, 1200);
  delay(120);
  noTone(BUZZER);

  delay(80);

  tone(BUZZER, 1800);
  delay(120);
  noTone(BUZZER);
}

void beepDenied() {

  for (int i = 0; i < 3; i++) {

    tone(BUZZER, 300);
    delay(120);
    noTone(BUZZER);

    delay(80);
  }
}
```

```
void beepError() {

  for (int i = 0; i < 2; i++) {

    tone(BUZZER, 150);
    delay(400);
    noTone(BUZZER);

    delay(150);
  }
}
```

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```
// ================ HOME SCREEN ================
void showHomeScreen() {

  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print(ROOM_NAME);

  lastUpdate = 0;
}

// ================ RTC ================
void setupRTC() {

  if (!rtc.begin()) {

    lcd.clear();
    lcd.print("RTC Failed");

    while (1);
  }

  // AUTO SET RTC IF BATTERY LOST
  if (rtc.lostPower()) {

    rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));
  }
}
```

```
// ================ CLOCK ================
void updateClock() {

  if (millis() - lastUpdate >= 1000) {

    lastUpdate = millis();

    DateTime now = rtc.now();

    char line[17];

    sprintf(line,
            "%02d:%02d:%02d",
            now.hour(),
            now.minute(),
            now.second());

    lcd.setCursor(0, 1);
    lcd.print("                ");

    lcd.setCursor(0, 1);
    lcd.print(line);
  }
}
```

```
// ================ DOOR ================
void openDoor() {

  digitalWrite(RELAY_PIN, UNLOCK_STATE);

  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print("Door Open");

  beepSuccess();

  delay(5000);

  digitalWrite(RELAY_PIN, LOCK_STATE);

  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print("Door Locked");

  delay(1000);

  showHomeScreen();
}
```

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```
// ================ WIFI ================
void connectWiFi() {

  if (WiFi.status() == WL_CONNECTED)
    return;

  // PREVENT RECONNECT SPAM
  if (millis() - lastWiFiRetry < wifiRetryInterval)
    return;

  lastWiFiRetry = millis();

  lcd.clear();
  lcd.print("Connecting WiFi");

  Serial.println("\nConnecting WiFi...");

  WiFi.mode(WIFI_STA);

  // IMPORTANT
  WiFi.setSleep(false);

  // REMOVE OLD CONNECTION
  WiFi.disconnect();

  delay(500);
```

```
  WiFi.begin(
    ssids[currentWiFi],
    passwords[currentWiFi]
  );

  int attempts = 0;

  while (WiFi.status() != WL_CONNECTED &&
         attempts < 20) {

    delay(500);

    Serial.print(".");

    attempts++;
  }

  if (WiFi.status() == WL_CONNECTED) {

    Serial.println("\nWiFi Connected!");

    lcd.clear();
    lcd.print("WiFi Connected");

    beepSuccess();

    delay(1000);
```

```
    showHomeScreen();

    return;
  }

  Serial.println("\nWiFi Failed");

  lcd.clear();
  lcd.print("WiFi Failed");

  beepError();

  // SWITCH TO NEXT WIFI
  currentWiFi++;

  if (currentWiFi >= wifiCount)
    currentWiFi = 0;
}
```

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```
// ================ RFID PROCESS ================
void sendRFIDScan(String uid) {

  beepSuccess();

  DateTime now = rtc.now();

  char timeStr[9];

  sprintf(timeStr,
          "%02d:%02d:%02d",
          now.hour(),
          now.minute(),
          now.second());

  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print("UID:");
  lcd.print(uid.substring(0, 8));

  lcd.setCursor(0, 1);
  lcd.print(timeStr);

  delay(1500);
```

```
  // CHECK WIFI
  if (WiFi.status() != WL_CONNECTED) {

    lcd.clear();
    lcd.print("WiFi Offline");

    beepError();

    delay(1500);

    showHomeScreen();

    return;
  }
```

```
  HTTPClient http;

  String payload = "{";
  payload += "\"deviceId\":\"" + String(DEVICE_ID) + "\",";
  payload += "\"uid\":\"" + uid + "\",";
  payload += "\"status\":\"success\"";
  payload += "}";

  Serial.println("\nSending Request...");
  Serial.println(payload);

  http.begin(SERVER_URL);

  http.addHeader("Content-Type", "application/json");

  int code = http.POST(payload);

  Serial.print("HTTP CODE: ");
  Serial.println(code);
```

```
  if (code <= 0) {

    lcd.clear();
    lcd.print("Server Error");

    beepError();

    delay(1500);

    showHomeScreen();

    http.end();

    return;
  }

  String response = http.getString();

  Serial.println(response);

  http.end();
```

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```
330   StaticJsonDocument<2048> doc;
331
332   DeserializationError error =
333     deserializeJson(doc, response);
334
335   if (error) {
336
337     lcd.clear();
338     lcd.print("Parse Error");
339
340     beepError();
341
342     delay(1500);
343
344     showHomeScreen();
345
346     return;
347   }
```

```
349   bool ok = doc["success"] | false;
350
351   String scanStatus =
352     doc["scan"]["status"] | "";
353
354   String analysis =
355     doc["analysis"]["status"] | "";
356
357   String action =
358     doc["attendance"]["action"] | "";
359
360   String name =
361     doc["user"]["full_name"] | "Unknown";
362
363   lcd.clear();
```

```
365   // ACCESS GRANTED
366   if (ok &&
367       scanStatus == "success" &&
368       (analysis == "on_time" ||
369        analysis == "late")) {
370
371     lcd.setCursor(0, 0);
372
373     if (action == "TIME-IN")
374       lcd.print("TIME IN");
375
376     else if (action == "TIME-OUT")
377       lcd.print("TIME OUT");
378
379     else
380       lcd.print("ACCESS OK");
381
382     lcd.setCursor(0, 1);
383     lcd.print(name.substring(0, 16));
384
385     delay(1500);
386
387     openDoor();
388
389     return;
390   }
```

```
392   // ACCESS DENIED STATES
393   else if (analysis == "wrong_room") {
394
395     lcd.print("Wrong Room");
396   }
397
398   else if (analysis == "outside_schedule") {
399
400     lcd.print("Wrong Schedule");
401   }
402
403   else if (analysis == "no_schedule") {
404
405     lcd.print("No Schedule");
406   }
407
408   else if (scanStatus == "not_registered") {
409
410     lcd.print("Card Unknown");
411   }
412
413   else {
414
415     lcd.print("Access Denied");
416   }
```

```
  beepDenied();

  delay(1500);

  showHomeScreen();
}
```

```
// ================ SETUP ================
void setup() {

  Serial.begin(115200);

  SPI.begin();

  rfid.PCD_Init();

  Wire.begin(21, 22);

  lcd.init();
  lcd.backlight();

  pinMode(BUZZER, OUTPUT);

  pinMode(RELAY_PIN, OUTPUT);

  digitalWrite(RELAY_PIN, LOCK_STATE);

  setupRTC();

  beepSuccess();

  connectWiFi();

  showHomeScreen();
}
```

```
// ================ LOOP ================
void loop() {

  // AUTO WIFI RECONNECT
  if (WiFi.status() != WL_CONNECTED) {

    connectWiFi();
  }

  // RFID SCAN
  if (rfid.PICC_IsNewCardPresent() &&
      rfid.PICC_ReadCardSerial()) {

    String uid = "";

    for (byte i = 0; i < rfid.uid.size; i++) {

      if (rfid.uid.uidByte[i] < 0x10)
        uid += "0";

      uid += String(
               rfid.uid.uidByte[i],
               HEX
             );
    }
```

```
    uid.toUpperCase();

    sendRFIDScan(uid);

    rfid.PICC_HaltA();
    rfid.PCD_StopCrypto1();
  }

  updateClock();

  delay(100);
}
```

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B.3. Networking Laboratory Code

```
#include <WiFi.h>
#include <WiFiClientSecure.h>
#include <HTTPClient.h>
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include "RTClib.h"
#include <LiquidCrystal_I2C.h>
#include <ArduinoJson.h>

// ================ WIFI ================
const char* ssids[] = {
  "BOOM",
  "GlobeAtHome_5DD78_2.4"
};

const char* passwords[] = {
  "@M4xRT&VDSL",
  "025C300D"
};

const int wifiCount = 2;
int currentWiFi = 0;

unsigned long lastWiFiRetry = 0;
const unsigned long wifiRetryInterval = 10000;

// ================ SERVER ================
const char* SERVER_URL =
"https://ferocious-penholder-blaming.ngrok-free.dev/api/iot/scans";
```

```
// ================ ROOM ================
const char* DEVICE_ID = "Networking-Lab";
const char* ROOM_NAME = "Networking Lab";

// ================ RELAY ================
#define LOCK_STATE LOW        // FIXED
#define UNLOCK_STATE HIGH     // FIXED

#define SS_PIN 5
#define RST_PIN 4

MFRC522 rfid(SS_PIN, RST_PIN);

RTC_DS3231 rtc;
LiquidCrystal_I2C lcd(0x27, 16, 2);

#define BUZZER 25
#define RELAY_PIN 26

unsigned long lastClockUpdate = 0;
```

```
// ================ BUZZER ================
void beepSuccess() {
  tone(BUZZER, 1200); delay(120); noTone(BUZZER);
  delay(80);
  tone(BUZZER, 1800); delay(120); noTone(BUZZER);
}

void beepDenied() {
  for (int i = 0; i < 3; i++) {
    tone(BUZZER, 300); delay(120); noTone(BUZZER);
    delay(80);
  }
}

void beepError() {
  for (int i = 0; i < 2; i++) {
    tone(BUZZER, 150); delay(300); noTone(BUZZER);
    delay(100);
  }
}

// ================ HOME SCREEN ================
void showHome() {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print(ROOM_NAME);
}
```

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```
// ================ RTC ================
void setupRTC() {
  if (!rtc.begin()) {
    lcd.clear();
    lcd.print("RTC Failed");
    while (1);
  }

  if (rtc.lostPower()) {
    rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));
  }
}
```

```
// ================ WIFI ================
void connectWiFi() {
  if (WiFi.status() == WL_CONNECTED) return;

  if (millis() - lastWiFiRetry < wifiRetryInterval) return;
  lastWiFiRetry = millis();

  lcd.clear();
  lcd.print("Connecting WiFi");

  WiFi.mode(WIFI_STA);
  WiFi.setSleep(false);
  WiFi.disconnect();
  delay(300);

  WiFi.begin(ssids[currentWiFi], passwords[currentWiFi]);

  int attempts = 0;
  while (WiFi.status() != WL_CONNECTED && attempts < 20) {
    delay(500);
    attempts++;
  }
```

```
  if (WiFi.status() == WL_CONNECTED) {
    lcd.clear();
    lcd.print("WiFi Connected");
    beepSuccess();
    delay(800);
    showHome();
  } else {
    beepError();
    currentWiFi = (currentWiFi + 1) % wifiCount;
    showHome();
  }
}
```

```
// ================ RFID RESET ================
void resetRFID() {
  rfid.PICC_HaltA();
  rfid.PCD_StopCrypto1();
  delay(50);
  rfid.PCD_Init();
}

// ================ CLOCK ================
void updateClock() {
  if (millis() - lastClockUpdate < 1000) return;
  lastClockUpdate = millis();

  DateTime now = rtc.now();

  char buf[17];
  sprintf(buf, "%02d:%02d:%02d", now.hour(), now.minute(), now.second());

  lcd.setCursor(0, 1);
  lcd.print("                ");
  lcd.setCursor(0, 1);
  lcd.print(buf);
}
```

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```
// ================ DOOR CONTROL ================
void openDoor() {

  digitalWrite(RELAY_PIN, UNLOCK_STATE); // unlock on success

  lcd.clear();
  lcd.print("Door Open");
  beepSuccess();

  delay(5000);

  digitalWrite(RELAY_PIN, LOCK_STATE); // Alock again

  lcd.clear();
  lcd.print("Door Locked");

  delay(800);
  showHome();
}
```

```
// ================ RFID SEND ================
void sendRFID(String uid) {

  beepSuccess();

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("UID:");
  lcd.print(uid.substring(0, 8));

  DateTime now = rtc.now();

  char timeStr[9];
  sprintf(timeStr, "%02d:%02d:%02d", now.hour(), now.minute(), now.second());

  lcd.setCursor(0, 1);
  lcd.print(timeStr);

  delay(800);

  if (WiFi.status() != WL_CONNECTED) {
    lcd.clear();
    lcd.print("WiFi Offline");
    beepError();
    delay(1000);
    showHome();
    resetRFID();
    return;
  }
```

```
  WiFiClientSecure client;
  client.setInsecure();
  client.setTimeout(8000);

  HTTPClient http;

  String payload =
    String("{\"deviceId\":\"") + DEVICE_ID +
    "\",\"uid\":\"" + uid +
    "\",\"status\":\"success\"}";

  http.setTimeout(8000);

  if (!http.begin(client, SERVER_URL)) {
    lcd.clear();
    lcd.print("Server Error");
    beepError();
    delay(1000);
    showHome();
    resetRFID();
    return;
  }

  http.addHeader("Content-Type", "application/json");

  int code = http.POST(payload);
```

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```
  if (code <= 0) {
    lcd.clear();
    lcd.print("Server Error");
    beepError();
    delay(1000);
    showHome();
    http.end();
    resetRFID();
    return;
  }

  String response = http.getString();
  http.end();

  StaticJsonDocument<2048> doc;
  if (deserializeJson(doc, response)) {
    lcd.clear();
    lcd.print("Parse Error");
    beepError();
    delay(1000);
    showHome();
    resetRFID();
    return;
  }
```

```
  bool ok = doc["success"] | false;
  String scanStatus = doc["scan"]["status"] | "";
  String analysis = doc["analysis"]["status"] | "";
  String action = doc["attendance"]["action"] | "";
  String name = doc["user"]["full_name"] | "Unknown";

  lcd.clear();

  if (ok && scanStatus == "success" &&
      (analysis == "on_time" || analysis == "late")) {

    if (action == "TIME-IN") {
      lcd.print("TIME IN");
      beepSuccess();
    }
    else if (action == "TIME-OUT") {
      lcd.print("TIME OUT");
      beepSuccess();
    }
    else {
      lcd.print("ACCESS OK");
      beepSuccess();
    }

    lcd.setCursor(0, 1);
    lcd.print(name.substring(0, 16));

    delay(1200);
    openDoor();
  }
```

```
  else {
    lcd.print("Access Denied");
    beepDenied();
    delay(1000);
    showHome();
  }

  resetRFID();
}
```

```
// ================ SETUP ================
void setup() {

  Serial.begin(115200);

  SPI.begin();
  rfid.PCD_Init();

  Wire.begin(21, 22);

  lcd.init();
  lcd.backlight();

  pinMode(BUZZER, OUTPUT);
  pinMode(RELAY_PIN, OUTPUT);

  digitalWrite(RELAY_PIN, LOCK_STATE);

  setupRTC();
  connectWiFi();
  showHome();
}
```

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```
// ================ LOOP ================
void loop() {

  if (WiFi.status() != WL_CONNECTED) connectWiFi();

  if (rfid.PICC_IsNewCardPresent() &&
      rfid.PICC_ReadCardSerial()) {

    String uid = "";

    for (byte i = 0; i < rfid.uid.size; i++) {
      if (rfid.uid.uidByte[i] < 0x10) uid += "0";
      uid += String(rfid.uid.uidByte[i], HEX);
    }

    uid.toUpperCase();

    sendRFID(uid);

    resetRFID();
  }

  updateClock();
}
```

Appendix C - Authors

C.1. Scrum Master

LEE JOSHUA REYES

PROJECT LEADER

+63 977 191 5826
reyesleejoshua29@gmail.com
Bacoor, Cavite Philippines

EDUCATION

2023 - Present
St. Dominic College of Asia
Bachelor of Science in Information Technology

2021 - 2023
Adamson University
Senior High School

TECHNICAL SKILLS

- HTML
- CSS
- JAVA
- SQL
- C Programming
- JAVASCRIPT
- C++
- TypeScript

CERTIFICATION

2024
Microsoft PowerPoint
Certiport

2024
Microsoft Word
Certiport

2025
Database
Certiport

2026
HTML and CSS
Certiport

TOOLS

- Git
- GitHub
- Visual Studio Code
- Android Studio
- React
- Node.js
- Bruno
- MongoDB
- Dart
- Express
- Flutter
- Arduino IDE
- Firebase
- Postman

PROJECT EXPERIENCE

"Smart Clearance System: AI-Assisted Multi-Office Faculty Clearance System"
SDCA, 2025
Developed a Faculty Clearance System that tracks the faculty's status on clearance compliances.

"Development of an AI-Assisted Faculty Scheduling and Requirements Management System"
SDCA, 2026 - Present
Developing a Faculty Management System with attendance monitoring, scheduling, and clearance compliance integrated into a web.

C.2. Frontend Developer

Russel Macaspac

Project Contributor

CONTACT

Phone: +63 956-415-8844
Email: russelaaron2004@gmail.com
Linkedin: www.linkedin.com/in/ramacaspac
Address: Imus, Cavite Philippines
Portfolio: russel-aaron-macaspac.github.io/portfolio

TECHNICAL SKILLS

Programming Languages

- HTML
- CSS
- JAVA
- SQL
- TypeScript
- JAVASCRIPT
- C Programming
- C++

Tools

App and Web Development

- Git
- GitHub
- Visual Studio Code
- Android Studio
- React
- Node.js
- MongoDB
- Express
- Flutter
- Next.js
- Dart
- Firebase
- Postman
- Bruno
- Arduino IDE

EDUCATION

St. Dominic College of Asia | 2023-Present (current)
Bachelor of Science in Information Technology

Adamson University | 2021-2023
Senior High School | STEM

CERTIFICATION

Microsoft Power Point | 2024
Certiport

Microsoft Word | 2024
Certiport

Database | 2025
Certiport

HTML and CSS | 2026
Certiport

RESEARCH & CAPSTONE PROJECTS

"Smart Clearance System: AI Assisted Multi-Office Faculty Clearance System"
Developed a web-based clearance management system designed to streamline faculty clearance processing across multiple offices within an institution. The system features AI-assisted validation and status tracking to improve efficiency, reduce manual processing, and enhance communication between departments.

"AI-Assisted Web-Based Faculty Scheduling & Requirements Management System"
a web-based system that automates faculty scheduling and manages academic requirements through AI-assisted features for improved efficiency and organization. The system helps streamline schedule allocation, track faculty documents and requirements, and reduce manual administrative workload.

C.3. Backend Developer

JASON PELAGIO

pelagiojason02@gmail.com | 09217009905 | Bacoor, Cavite, Philippines

OBJECTIVE

Motivated and detail-oriented 3rd Year BSIT student seeking an OJT / internship opportunity to apply academic knowledge in real-world environments. Eager to contribute to technology-driven projects, gain hands-on industry experience, and grow as a developer with a particular interest in AI integration and web-based systems.

EDUCATION

St. Dominic College of Asia 2022 – Present
Bachelor of Science in Information Technology
Bacoor, Cavite, Philippines

TECHNICAL SKILLS

- Programming & Development: Gained through coursework and project work in BSIT
- Artificial Intelligence Integration: Developed and implemented an AI model for open-ended question synthesis
- Web Development: Exposure to front-end and back-end concepts through academic projects
- Database Management: Foundational knowledge in database design acquired through IT coursework
- Systems Analysis & Design: Applied in ongoing faculty scheduling and requirements management system project

ACADEMIC PROJECTS

AI-Powered School Feedback Synthesis Tool *2024*
An academic project focused on developing an AI model that synthesizes student open-ended responses and evaluates emotional/sentiment metrics based on their answers. The core AI component was successfully completed capable of processing open-ended questions, generating meaningful insights, and exporting structured results as JSON and PDF reports. The project concluded at the AI development phase.

- Built and trained an AI model to analyze and synthesize qualitative student responses
- Implemented JSON parsing and automated PDF generation from AI output
- Applied sentiment analysis techniques to measure student emotional engagement

AI-Assisted Web-Based Faculty Scheduling & Requirements Management System *2024 – Present*
An ongoing capstone-level development project aimed at streamlining faculty scheduling and academic requirements management in a school setting. The system incorporates AI assistance to optimize scheduling logic and reduce conflicts. Currently in active development as part of academic requirements.

- Contributing to the design and development of an AI-assisted scheduling algorithm
- Involved in requirements gathering, system design, and web-based implementation
- Collaborating with a team to build a full-stack web application

RELEVANT COURSEWORK

- Web Development (Front-End & Back-End)
- Database Management Systems
- Information Management
- Computer Networks

C.4. System Analyst

David Simon Fauni

IT Specialist

CONTACT

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Email: simonfauni08@gmail.com
Address: Dasma., Cavite
Portfolio: https://david-simon-fauni.github.io/portfolio/

TECHNICAL SKILLS

Programming Languages

- HTML5
- CSS3
- JAVA
- JAVASCRIPT
- C Programming

Tools

App and Web Development

- Git
- GitHub
- Visual Studio Code
- Android Studio

EDUCATION

St. Dominic College of Asia | 2023-2025 (current)
Bachelor of Science in Information Technology

Adamson University | 2021-2023
Senior High School | STEM

CERTIFICATION

Microsoft Power Point | 2024
Certiport

Microsoft Word | 2024
Certiport

Database | 2025
Certiport